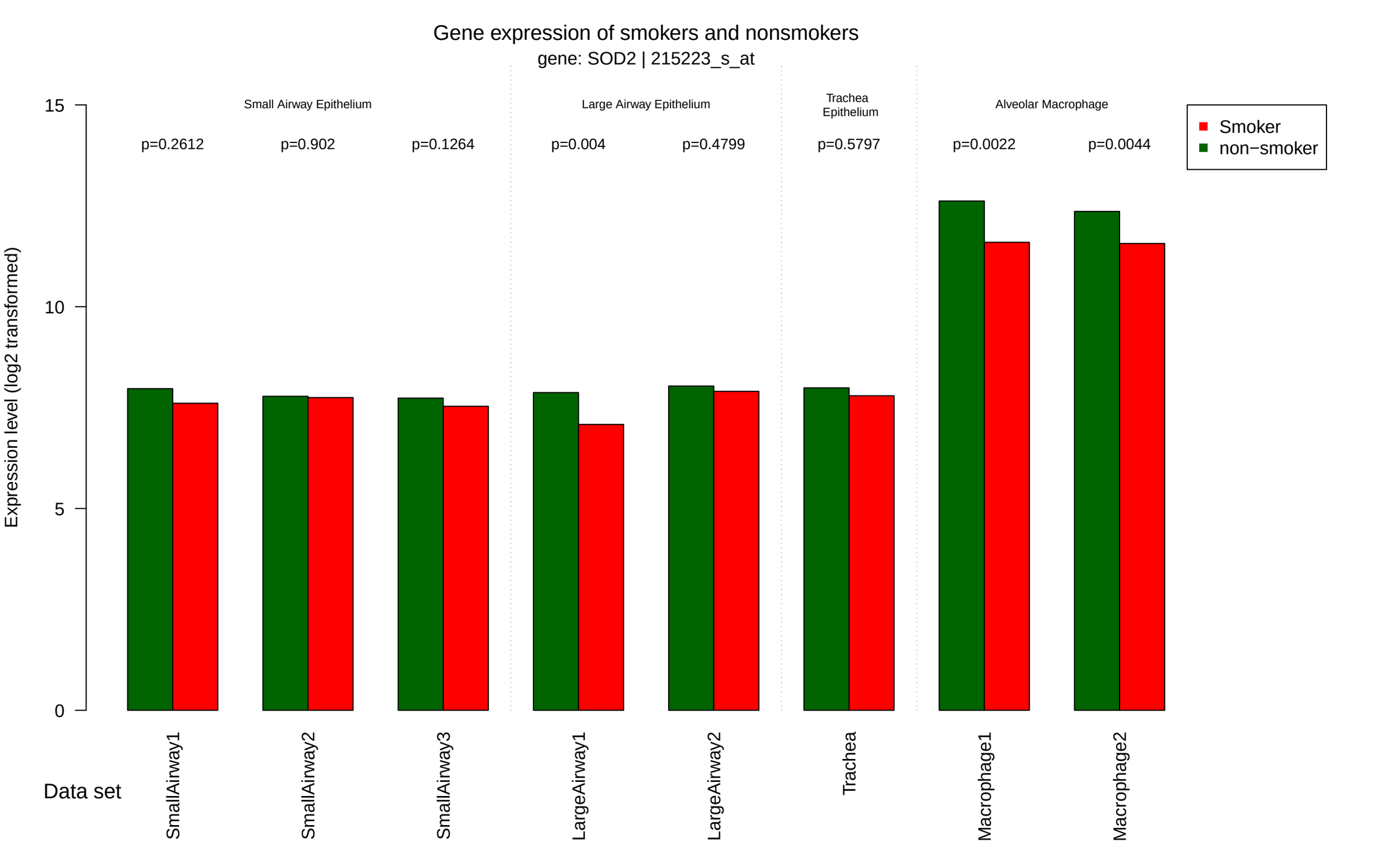
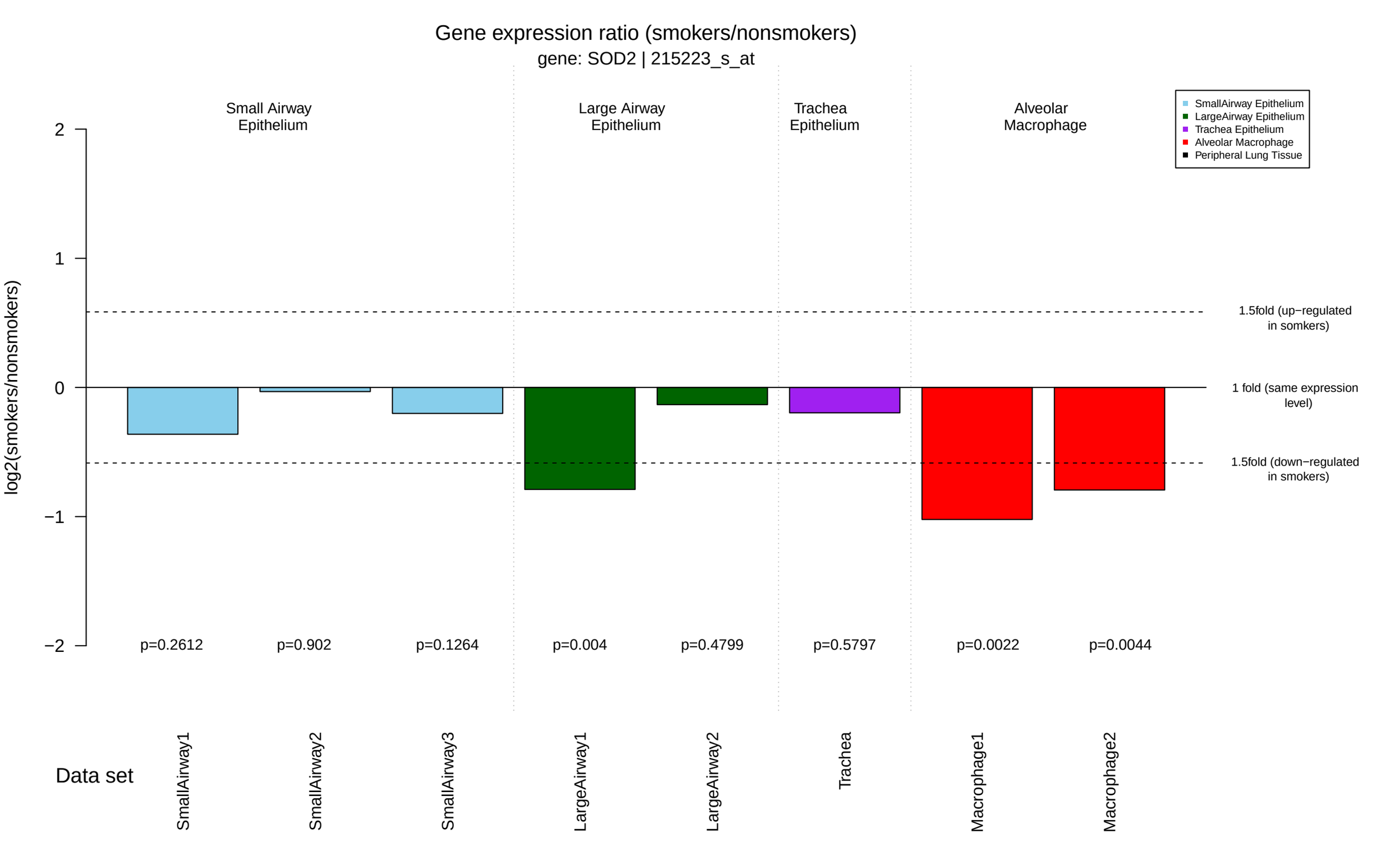
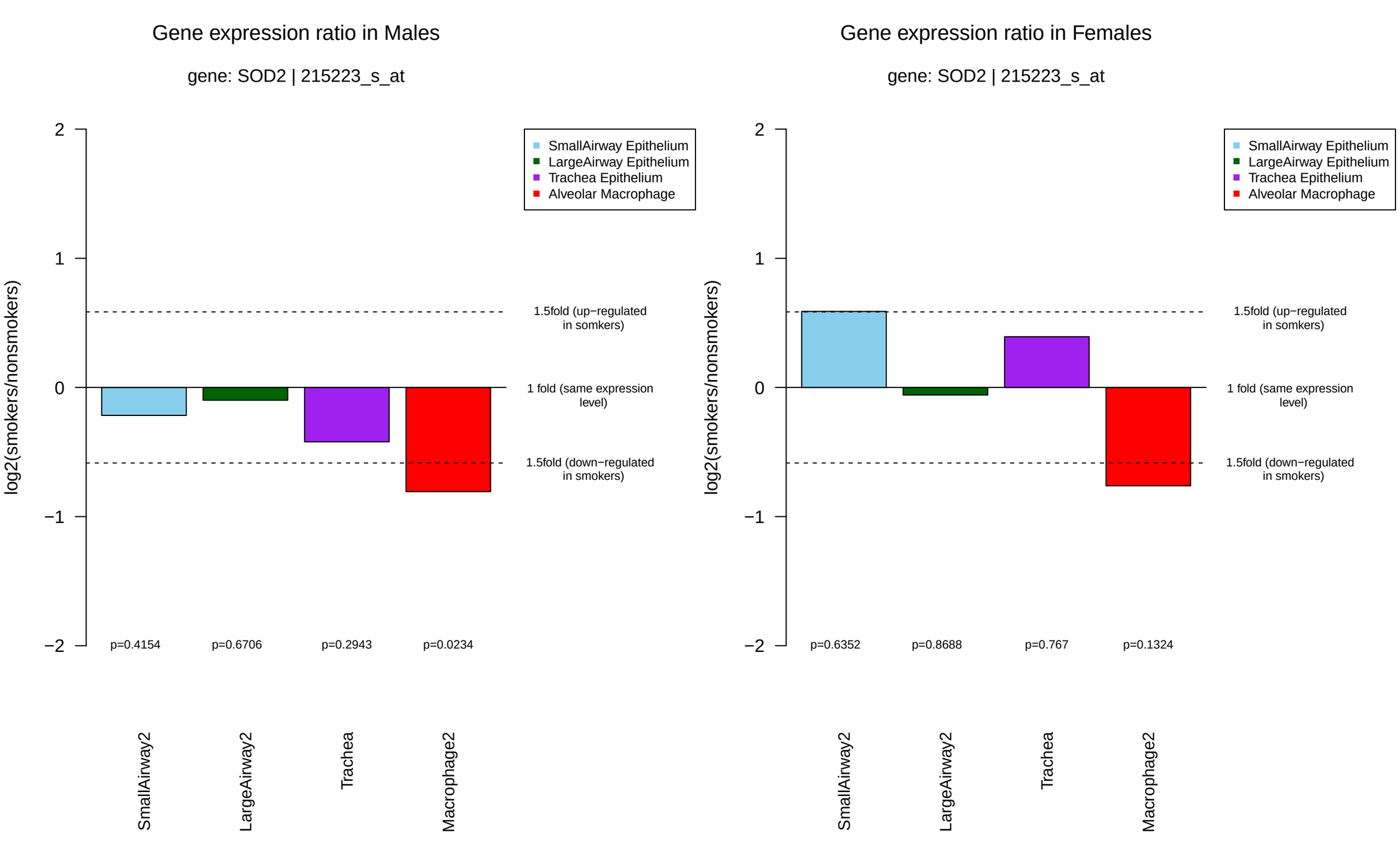
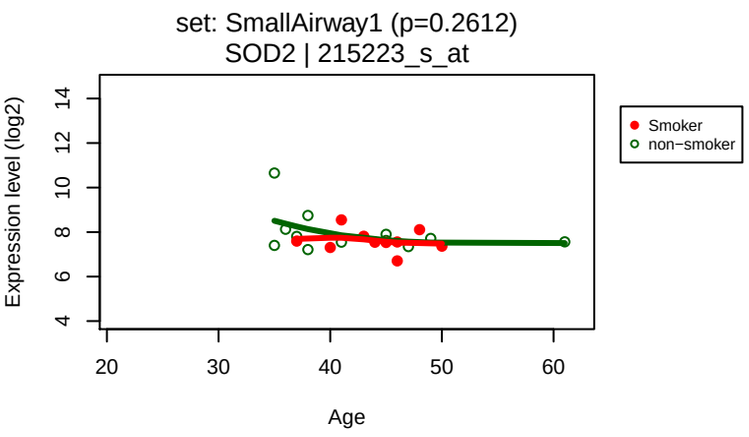
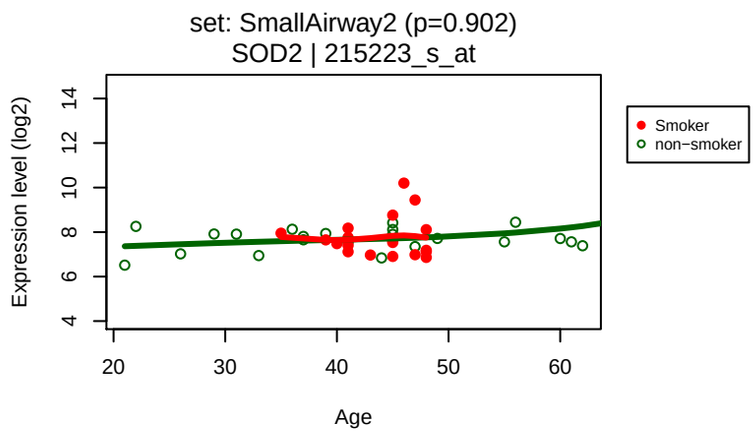
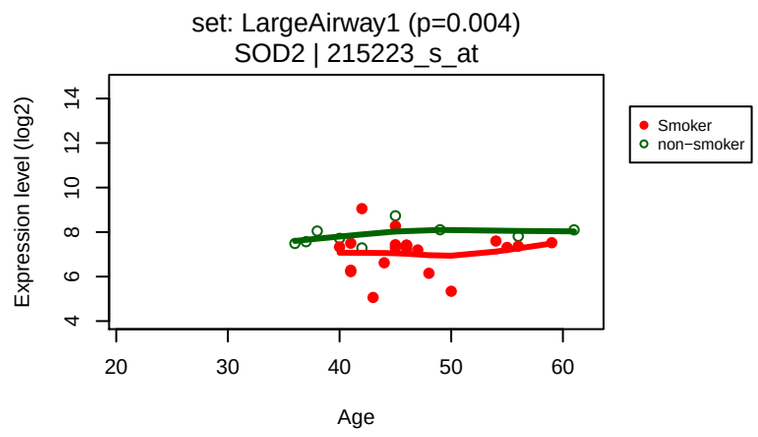

Gene expression of smokers and nonsmokers
gene: SOD2 | 215223_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2612
p=0.902
p=0.1264
p=0.004
p=0.4799
p=0.5797
p=0.0022
p=0.0044
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

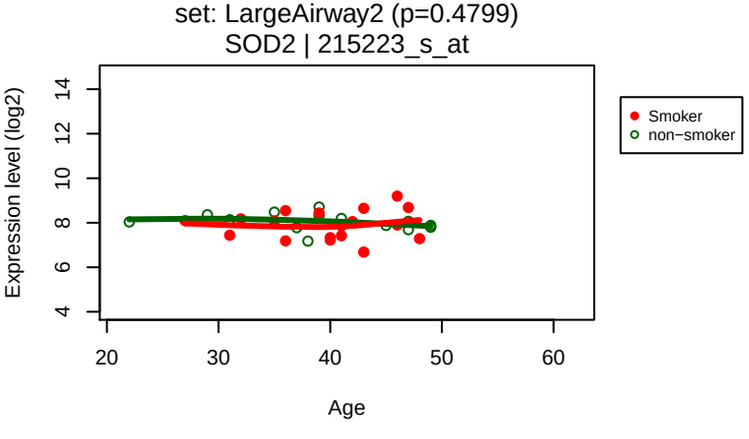

Gene expression ratio (smokers/nonsmokers)
gene: SOD2 | 215223_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2612
p=0.902
p=0.1264
p=0.004
p=0.4799
p=0.5797
p=0.0022
p=0.0044
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

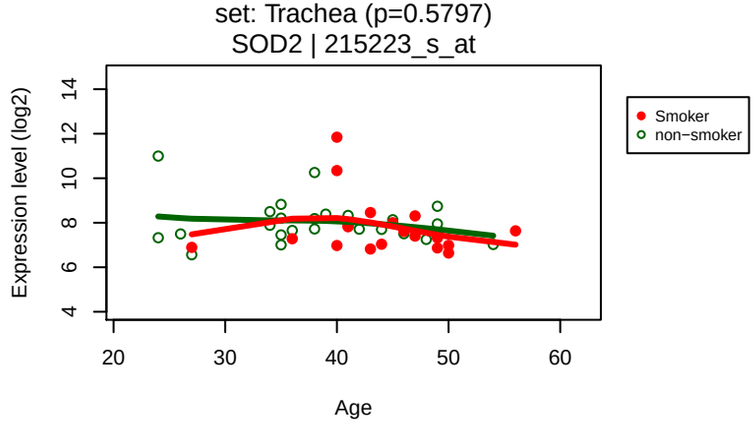

Gene expression ratio in Males
gene: SOD2 | 215223_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4154
p=0.6706
p=0.2943
p=0.0234
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOD2 | 215223_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6352
p=0.8688
p=0.767
p=0.1324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

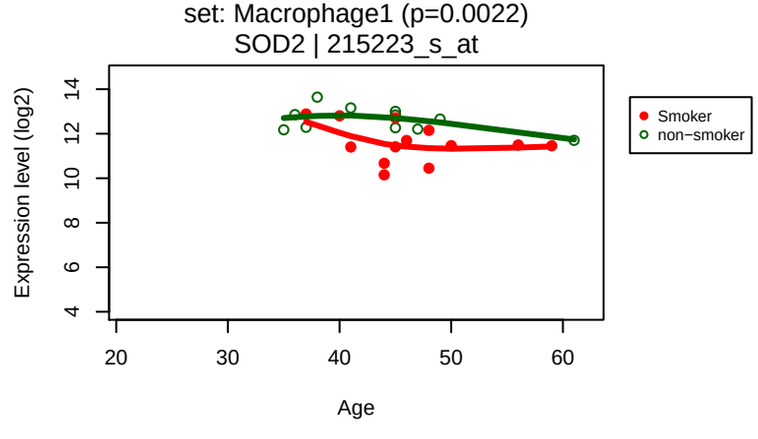

set: SmallAirway1 (p=0.2612)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker

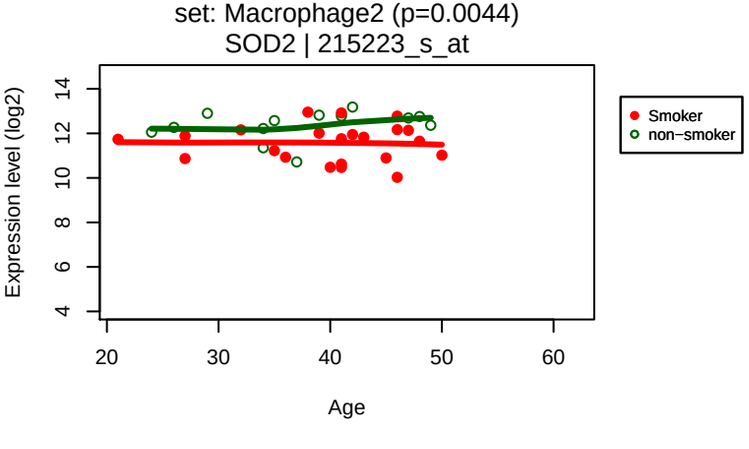

set: SmallAirway2 (p=0.902)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker

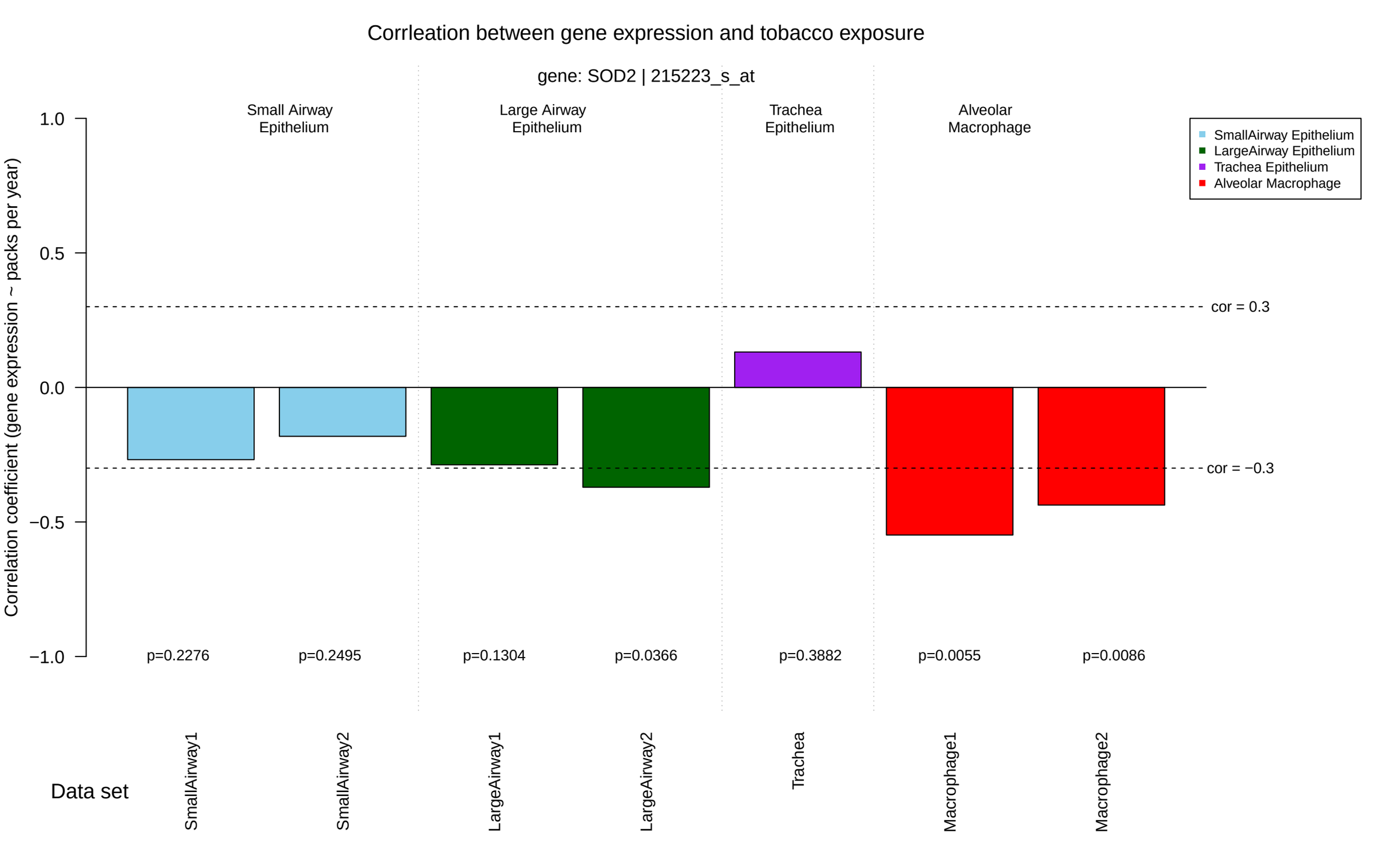

set: LargeAirway1 (p=0.004)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4799)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5797)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0022)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0044)
SOD2 | 215223_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOD2 | 215223_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2276
p=0.2495
p=0.1304
p=0.0366
p=0.3882
p=0.0055
p=0.0086
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2